First Order Logic

Create a knowledge base consisting of first order logic statements and prove the given query using forward reasoning.

Pseudocode:

START

1. Initialize Knowledge Base (KB) with facts and rules.

2. Repeat until no new facts can be derived:

   a. Set NEW = empty.

   b. For each rule in KB:
      i. Match the conditions of the rule
         with the existing facts.
      ii. Apply variable substitution.
      iii. If all conditions are satisfied,
           derive the conclusion.
      iv. Add the new conclusion to NEW
          if it is not already in KB.

   c. Add NEW facts to KB.

3. Check whether the query Criminal(Robert)
   exists in KB.

4. If the query exists:
      Print "Robert is Criminal".
   Otherwise:
      Print "Query cannot be proved".

STOP

Code:

```
def is_variable(x):
    return x.startswith("?")

def unify(pattern, fact, substitution):
    substitution = substitution.copy()
```

```
    for p, f in zip(pattern, fact):
        if is_variable(p):
            if p in substitution:
                if substitution[p] != f:
                    return None
            else:
                substitution[p] = f
        elif p != f:
            return None

    return substitution


def match_conditions(conditions, facts, substitution=None):
    if substitution is None:
        substitution = {}

    if not conditions:
        return [substitution]

    results = []

    first = conditions[0]
    remaining = conditions[1:]

    for fact in facts:
        new_sub = unify(first, fact, substitution)

        if new_sub is not None:
            results.extend(
                match_conditions(remaining, facts, new_sub)
            )

    return results


# Initial facts
facts = [
    ("Owns", "A", "T1"),
    ("Missile", "T1"),
    ("American", "Robert"),
    ("Enemy", "A", "America")
]

# Rules: conditions -> conclusion
rules = [
```

```
    (
        [("Missile", "?x")],
        ("Weapon", "?x")
    ),
    (
        [("Missile", "?x"), ("Owns", "A", "?x")],
        ("Sells", "Robert", "?x", "A")
    ),
    (
        [("Enemy", "?x", "America")],
        ("Hostile", "?x")
    ),
    (
        [
            ("American", "?p"),
            ("Weapon", "?q"),
            ("Sells", "?p", "?q", "?r"),
            ("Hostile", "?r")
        ],
        ("Criminal", "?p")
    )
]

print("Initial Knowledge Base:")
for fact in facts:
    print(fact)

# Forward Chaining
changed = True

while changed:
    changed = False

    for conditions, conclusion in rules:
        substitutions = match_conditions(
            conditions, facts
        )

        for sub in substitutions:
            new_fact = tuple(
                sub.get(x, x) for x in conclusion
            )

            if new_fact not in facts:
                facts.append(new_fact)
                changed = True
```

```
                print("Derived:", new_fact)

# Check the query
query = ("Criminal", "Robert")

print("\nFinal Result:")

if query in facts:
    print("Query proved successfully!")
    print("Robert is Criminal.")
else:
    print("Query cannot be proved.")
```

Output:

